Huawei Access Node Optical Transport Failure SOP (BAS-ACC-53)

Node: BAS-ACC-53 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5576

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for addressing Optical Transport Failures on the Huawei Access Node BAS-ACC-53 located at the Basavanagudi POP. The Optical Transport Failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, which can significantly impact the reliability and uptime of the network. In this context, the failure domain occurs on the Access Role, which is responsible for providing connectivity between the customer's premises and the service provider's network. The Huawei NE40E platform, used in this scenario, employs a robust Optical Transport Network (OTN) architecture to ensure high-speed data transmission over fiber-optic cables. However, SFP/XFP degradation, fiber cuts, or Rx/Tx power loss can compromise the OTN's integrity, leading to service disruptions. This SOP provides a structured approach to diagnose and resolve such failures, ensuring minimal downtime and maximizing network availability.

Pre-requisite Checks

- Pre-requisite: Verify that the Basavanagudi POP is accessible and that remote hands are dispatched to the site., Status: Confirmed

- Pre-requisite: Check the status of redundant links to determine if they are operational and can take over in case of a failure., Status: Confirmed

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to excessive temperature fluctuations or optical power imbalances.	High
Fiber cuts resulting from external damage or maintenance activities.	Medium
Rx/Tx power loss due to attenuation or signal degradation over long fiber spans	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-fiber link all, Expected Output: Optical fiber link information: ..., Description: This command displays a list of optical fiber links, including their ID, wavelength, and state. It helps identify potential issues with fiber cuts or SFP/XFP degradation.

Command: display optical-fiber link fault, Expected Output: Optical fiber link fault information: ..., Description: This command provides detailed information about fiber link faults, including the fault location and type. It assists in isolating the root cause of the failure.
```

```
Command: display transceiver all, Expected Output: Transceiver
information: ..., Description: This command displays a list of
transceivers (SFP/XFP) installed on the node, including their type,
status, and optical power levels. It helps identify issues with
SFP/XFP degradation or misconfiguration.
```

Resolution Steps

Step 1: Step: Replace faulty SFP/XFP module, Command: replace transceiver 1/1/1, Description: Replace the faulty SFP/XFP module with a new one to restore the fiber link's integrity.

Step 2: Step: Repair or replace fiber cable, Command: repair optical-fiber link 1/1/1, Description: Repair or replace the damaged fiber cable to restore the fiber link's integrity and maintain network availability.

Step 3: Step: Adjust optical power levels, Command: set optical-fiber link 1/1/1 power-level high, Description: Adjust the optical power levels to ensure the fiber link operates within the recommended range, preventing signal degradation or attenuation.

Post-Check Verification

```
Command: display optical-fiber link all, Expected Output: Optical
fiber link information: ..., Description: Verify that the fiber link
is operational and that the SFP/XFP module is functioning correctly.

Command: display transceiver all, Expected Output: Transceiver
information: ..., Description: Verify that the SFP/XFP module is
installed and functioning correctly, with appropriate optical power
levels.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: replace transceiver 1/1/1, Description: Restore the original SFP/XFP module to its previous state to maintain network consistency.

- Step: Restore original fiber cable, Command: restore optical-fiber link 1/1/1, Description: Restore the original fiber cable to its previous state to maintain network consistency.